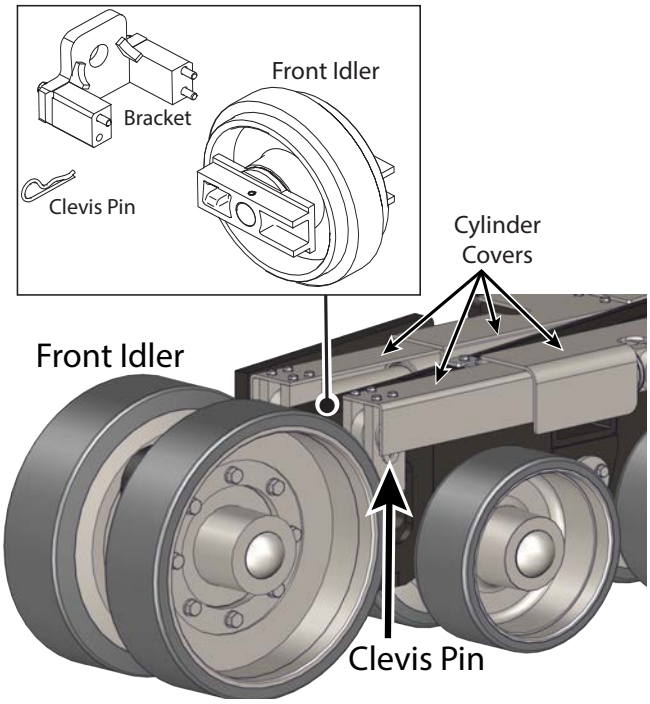


Figure 5-35. Front Idler

Front Idler

DANGER Crush hazard! Always use safety blocks or jack stands when working under the paver to prevent serious injury or even death.

Follow this procedure for replacing the front idler:

1. Follow the procedure for removing the track on ***Pages 5-34 to 5-35.***
2. Remove the front cylinder covers. **(Figure 5-35)**
3. Remove the front idler.
4. Install new idler, ensuring cylinder and clevis pin are aligned properly. Reattach front cylinder covers and track.

Two-Speed Hydraulic Motors

Move paver onto a level ramp where you can access the undercarriage.

DANGER Crush hazard! Always use safety blocks in addition to jack when working under the paver to prevent serious injury or even death.

1. Disconnect, label and plug fittings on hydraulic hoses to the hydraulic motors. **(Figure 5-36)**

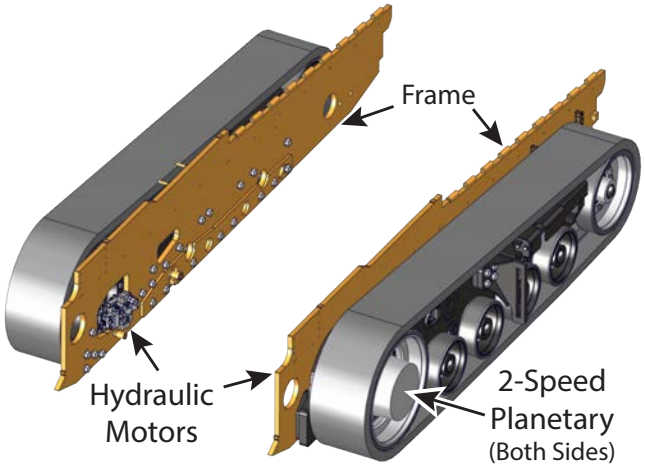


Figure 5-36. Hydraulic Motors and Torque Hubs

2. Place blocks sufficient to support the hydraulic motors underneath them.
3. Remove the hardware attaching the hydraulic motor to the planetary torque hub and carefully separate motor from the planetary. **(Figure 5-37)**

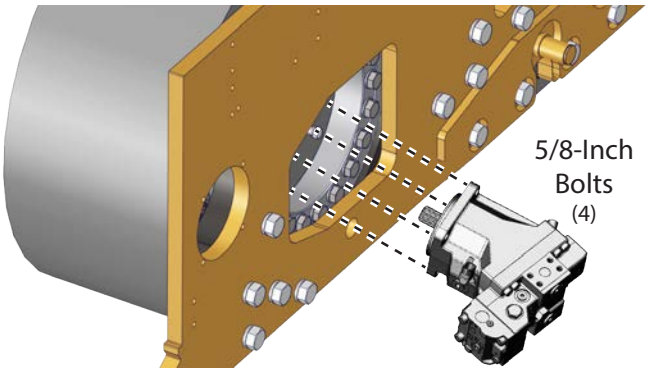


Figure 5-37. Remove Hydraulic Motor

4. Remove the O-Ring and drain hydraulic oil from the motor. Discard or repair the motor as appropriate.
5. Lubricate a new O-ring with hydraulic oil and install onto the planetary.
6. Reattach new or rebuilt hydraulic motor to planetary and refer to ***Torque Tables in* Section 2** to torque bolts. Reconnect hydraulic hoses.
7. Start the engine and let machine run for 10 minutes to work air out of the system. Check for any leaks.

Planetary Torque Hub

If the planetary torque hub has been damaged or worn, use the following procedures for replacement.

⚠ DANGER Crush hazard! Always use safety blocks in addition to jack when working under the paver to prevent serious injury or even death.

1. Remove the track by following **Steps 1 - 6** on ***Page 5-34.***
2. Remove all bolts around the hydraulic motor, then remove the four bolts holding the hydraulic motor and carefully separate motor. **(Figure 5-38)**

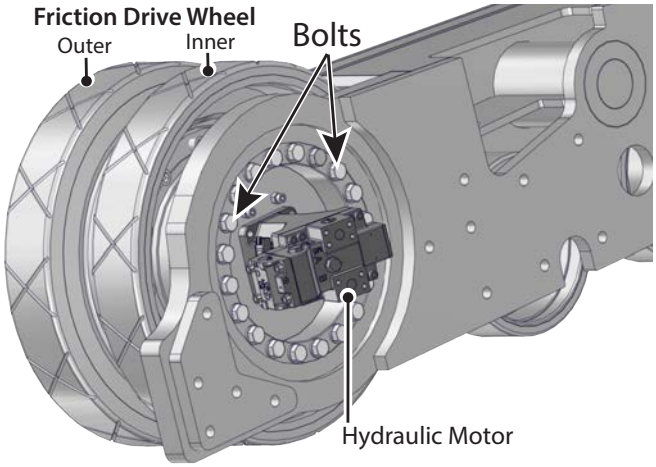


Figure 5-38. Inner Friction Drive Wheel Bolts

3. Inspect the O-ring on the hydraulic motor. Replace if worn, described in the next section.
4. Remove the inner and outer friction drive wheels, then remove the torque hub. **(Figure 5-39)**

NOTE: Mark location of the torque hub to the undercarriage before removing to ensure the planetary is reinstalled in the same position.

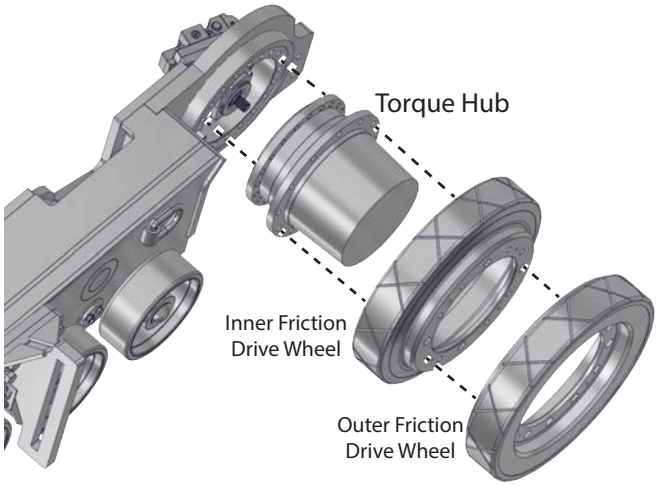


Figure 5-39. Friction Drive Wheel and Torque Hub

5. Install new torque hub into position to align with the hydraulic motor. Place thread-locking adhesive on hub bolts and torque. ***(See Torque Tables in* Section 2**).
6. Reinstall components in reverse order.
7. Lower paver to the ground.
8. Fill hub with oil. ***(Page 5-10)***
9. Start paver and check for any leaks with engine running.

NOTICE DO NOT use your bare hands when checking for hydraulic leaks as serious injury can result from an oil leak under high pressure. Always wear safety glasses and protective clothing.

⚠ WARNING Stop the engine immediately if any hydraulic leak is noted. DO NOT start the engine until the leak has been repaired.

5

REPLACEMENT PROCEDURES

Gear Pumps

Hydraulic fluid is transmitted throughout the machine to motors and cylinders, becoming pressurized according to the resistance present. The fluid is controlled by valves and distributed through hoses and tubes. One load-sense pump, one tandem variable volume drive pump, and two auxiliary pumps play a significant role in producing hydraulic power by supplying hydraulic fluid to system components: **(Figure 5-40)**

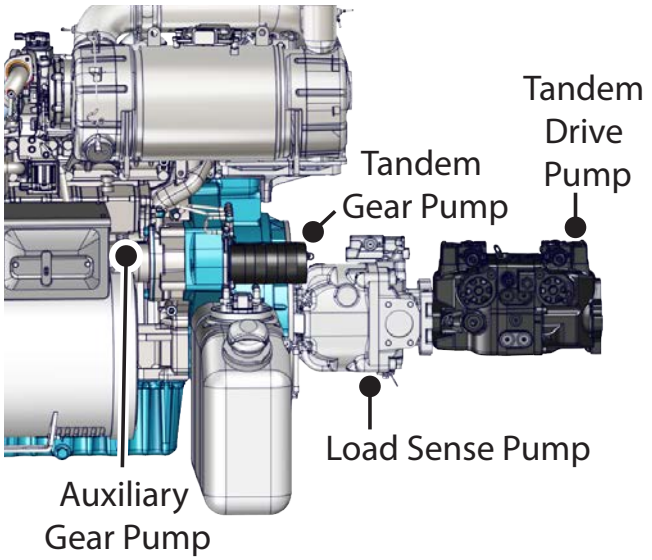


Figure 5-40. Gear Pumps

The tandem drive, load-sense, and rear auxiliary pumps are accessed via the right access panel. The tandem drive pump is attached to the load-sense pump. **(Figure 5-40).** The front auxiliary gear pump is easier to access from the top by raising the hood.

⚠CAUTION **The pump assemblies are very heavy and must be properly supported with a sling before continuing the following steps. DO NOT loosen mounting bolts unless the pump is sufficiently supported.**

1. Identify which pump needs to be replaced.
2. Label and disconnect the hoses to the pump slated for replacement, capping the hoses and fittings on the pump and hoses. Disconnect any wires to pump if applicable.
3. Remove any hardware attaching the pump.
4. Using a sling, carefully lift the pump assembly out of the machine and place on a flat surface.
5. Inspect the O-ring(s) on end of the pump for wear or damage. Replace if needed. Put a small amount of hydraulic oil onto the O-Ring and reinstall.
   - If replacing the load sense pump, check the drive coupling for wear or damage. Remove all bolts and pull out the coupling from the flywheel. Clean the flywheel and threaded bolt holes with ether, then install the new drive coupling into the flywheel if needed.
6. Align the new pump with the mounting holes and reattach pump, covering bolts with thread-locking adhesive. **(Figure 5-41)**

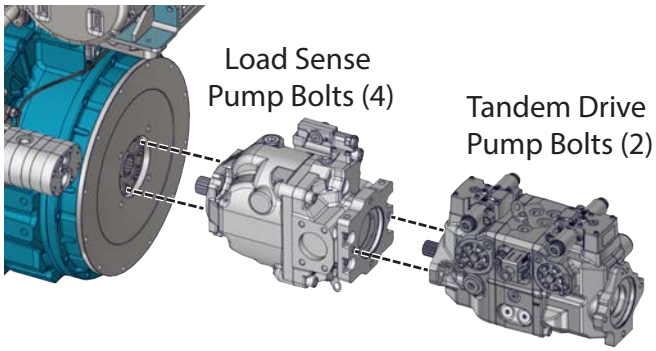


Figure 5-41. Hydraulic Pump Assemblies

7. Refer to ***Torque Tables in*** **Section 2** and torque bolts.
8. Remove plug caps and reconnect hydraulic hoses to pump as previously labeled.
9. Check hydraulic oil level. Add more if needed.
10. Turn the ignition key to Start.
11. Press and hold the Start/Stop button on the main control panel for five (5) seconds and release. Wait another five (5) seconds and repeat several times for one minute.
12. With the engine running, check for leaks.
13. If no leaks are noted, let machine idle approximately 10 minutes to allow the pump and motor cases to fill with hydraulic oil before operating.

Hydraulic Pump/Motor Start-Up

Pre-Start

If the hydraulic system is down as a result of a major component failure:

1. Drain and clean the tank and system components (hoses, valves, fittings and cooler) to ensure it is free from metallic debris and other contamination. Failure to do so may result in damage to the pump(s) and/or other components on start-up.
2. Change all the filters.
3. Change the fluid. On large systems where the cost of changing the fluid may be prohibitive, the fluid should be flushed until a cleanliness level of ISO 4406 18/16/13 or better is achieved.

Installation and Start-Up

When installing hydraulic motors, it is important that the mounting flange of the motor makes full contact with the mounting surface of the application. Mounting hardware of the appropriate grade and size must be used:

- Torque the bolts to the wet torque value defined for the capscrew grade and size. ***(See Section 2 Torque Tables.)***
- Apply Loctite 271 to the bolt threads.

⚠CAUTION **It is CRITICAL to use the correct mounting hardware.**

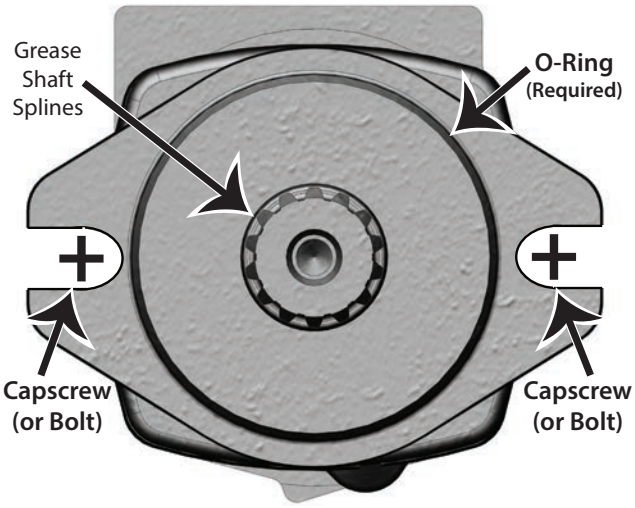


Figure 5-42. Hydraulic Motor Mount

Hubs, pulleys, sprockets and couplings must be properly aligned to avoid inducing excessive thrust or radial loads. Although the output device must fit the shaft tightly, a hammer should never be used to install any type of output device onto the shaft. The port plugs should only be removed from the motor when the system connections are ready to be made.

To avoid contamination:

- Remove all matter from around the ports of the motor and the threads of fittings.
- Fill the case of the motor being used (piston-type, gear motor, gerotor motor, geroller motor or vane motor) with clean hydraulic fluid through the highest case drain port and connect the case drain line. Failure to do so will result in damage to the motor through inadequate lubrication on start-up.
- Units that are mounted vertically with the shaft up require special attention to ensure the fluid level in the case is high enough to lubricate the front shaft bearing(s).

Once all system connections are made, the motor should be run for 15 - 30 minutes at no-load and half-speed to bleed air from the hydraulic system.

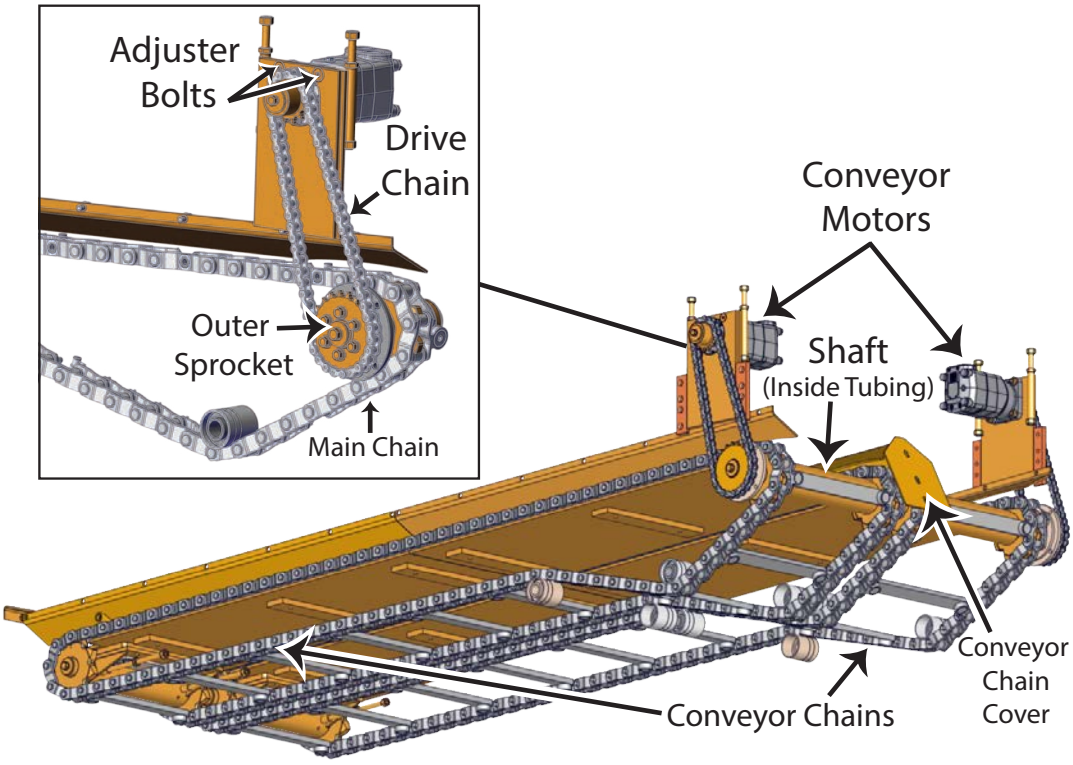


Figure 5-43. Conveyor Components

Rear Conveyor Shaft, Bearings and Sprockets

1. Raise the front lip at front of conveyor.
2. Relieve the chain tension.
3. Remove the rear conveyor chain cover. **(Figure 5-43)**
4. Remove the drive chain.
5. Remove the bolt and washers from the outer sprocket. **(Figures 5-44 and 5-45)**

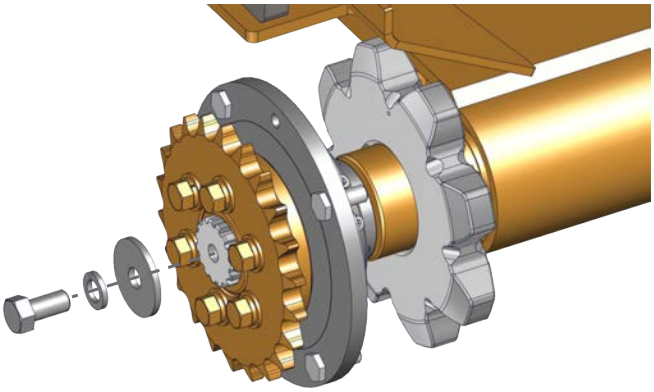


Figure 5-44. Remove Outer Bolt and Washers

6. Remove the remaining six bolts and washers. **(Figure 5-45)**
7. Remove the outer sprocket weldment.
8. Remove the conveyor drive coupling weldment.

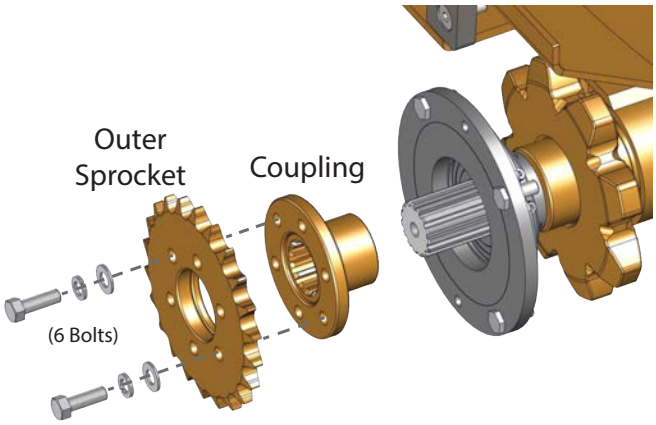


Figure 5-45. Remove Outer Sprocket and Coupling

9. Loosen and remove set screws from the outer drive sprocket bearing, then remove. **(Figure 5-46)**
10. Remove the outer conveyor sprocket bearing and mount plate (attached to the frame).
11. Remove the outer conveyor sprocket weldment.
12. Remove the hardware from the inner side of the shaft, then remove shaft.

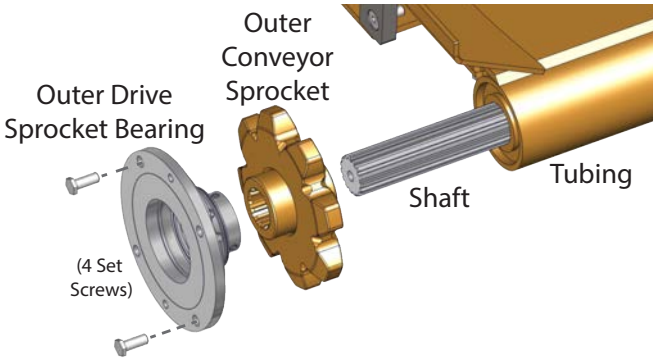


Figure 5-46. Remove Outer Drive Sprocket Weldment

13. Remove the hardware from the inner conveyor sprocket on other side of the shaft, then remove shaft. **(Figure 5-47)**
14. Remove outer conveyor bearings.
15. If replacing inner conveyor bearings and/or sprocket, remove using the same procedure as **Steps 5 - 11.**

NOTE: Rear inside tubing can be removed for changing the inner bearings.

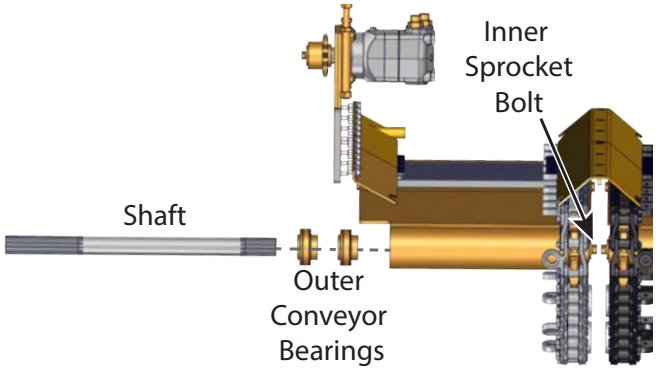


Figure 5-47. Remove Outer Sprocket and Coupling

16. Reinstall components in reverse order, replacing worn or damaged components with new ones, ensuring proper alignment.

NOTE: The outer conveyor sprocket must be aligned with the inner sprocket.

Conveyor Drive Motors

1. Relieve the conveyor chain tension. **(Figure 5-43)**
2. Remove the drive chain.
3. Disconnect hydraulic hoses from conveyor motor.
4. Remove four nuts and cap screws holding hydraulic motor to the slide mount. **(Figure 5-48)**

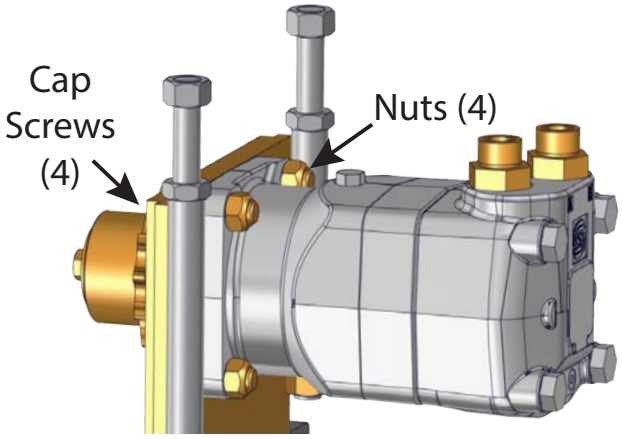


Figure 5-48. Remove Conveyor Drive Motor

5. Carefully slide the conveyor drive motor out of slide mount.
6. Slide new or repaired drive motor into slide mount.
7. Secure with cap screws and nuts.
8. Reconnect hydraulic hoses to motor.
9. Reinstall the drive chain and readjust conveyor chain tension.
10. Check for leaks.

5



Figure 5-49. Hopper Wing Cylinders

Front Hopper Wing Cylinders

Use the following procedure to replace the front hopper wing cylinder(s). **(Figure 5-49)**

1. Retract hopper wings completely.
2. Secure closed wing so it cannot fall open after cylinder is disconnected.
3. Disconnect hydraulic hose.
4. Remove clevis pin from the top cylinder mount and bolt from the bottom cylinder mount.
5. Slide cylinder off.
6. Slide new cylinder onto the cylinder mount.
7. Reinstall bolt.
8. Clean hydraulic hoses and connect to the cylinder.
9. Extend cylinder rod so that the hole in rod lines up with upper mount.
10. Attach clevis pin.

Rear Hopper Wing Cylinders

Use the following procedure to replace the rear hopper wing cylinder(s). **(Figure 5-49)**

1. Follow **Steps 1 - 2** in the previous section.
2. Raise the hood to access the bolt cover at the rod end of the cylinder.
3. Remove top bolt.
4. Fold out cylinder and disconnect hoses.
5. Disconnect the cylinder mount and slide off cylinder.
6. Slide new cylinder onto the mount.
7. Cover bolts with thread-locking adhesive before attaching new cylinder back onto the mount, then torque bolts ***(see Section 2).***
8. Clean the hydraulic hoses and reattach.
9. Extend the cylinder so that the bolt at the rod end lines up with the mount.
10. Cover the mount bolt with thread-locking adhesive and tighten.
11. Reattach bolt cover and close hood.

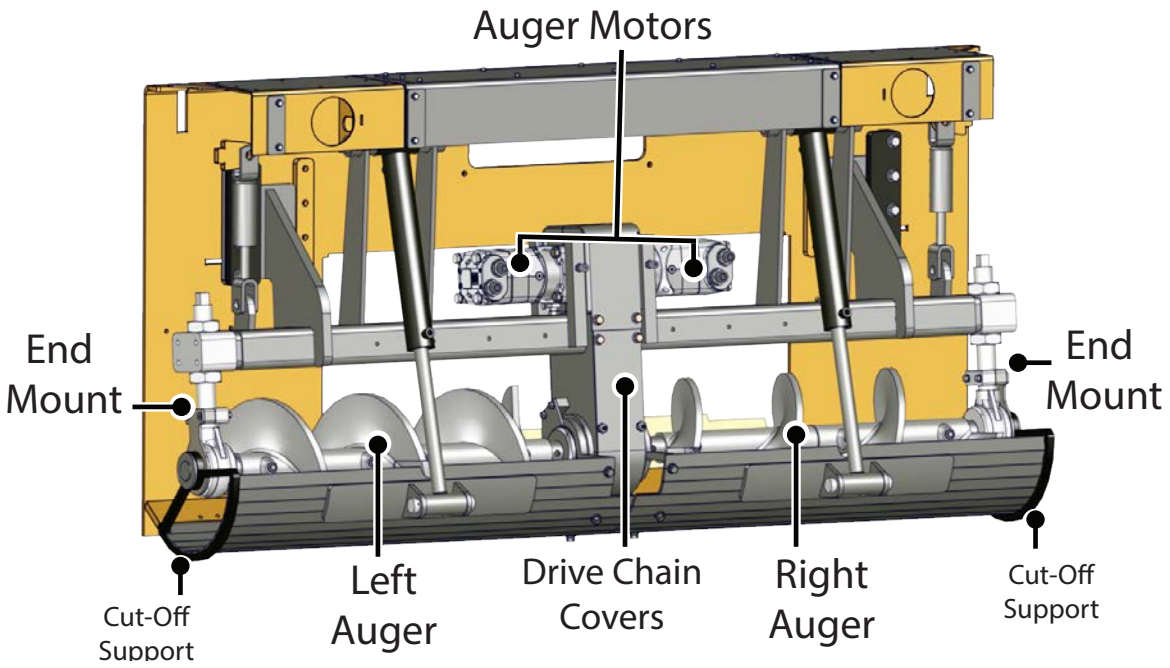


Figure 5-50. Auger Components

Auger Motors

1. Extend screed extension completely.
2. Clean asphalt buildup around the motor(s).
3. Disconnect and label the hydraulic hoses. Plug fittings.
4. Loosen the auger tension adjuster(s). **(Figure 5-51)**

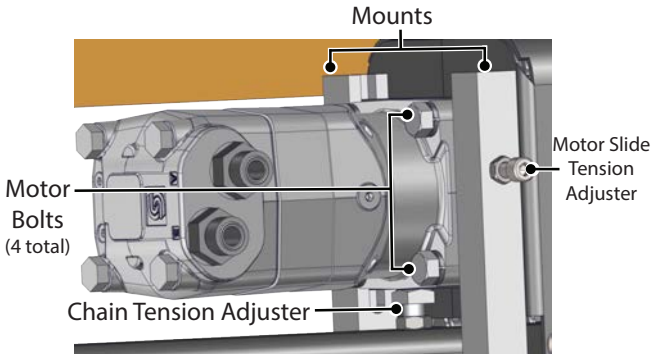


Figure 5-51. Auger Motor Replacement

5. Remove the cap screw on each side of the motor mount and remove mount.
6. Remove the four motor bolts and washers, and carefully slide out the auger motor.
7. Align and reattach the motor mounts.
8. Cover motor bolts with thread-locking adhesive, reinsert and tighten.
9. Align and reattach the motor mounts.
10. Retighten the tension adjuster to achieve 1/4 inches of auger chain slack.
11. Lubricate with chain lube.
12. Reconnect the hydraulic hoses and check for leaks.

Auger End Mounts

1. Extend the screed extensions completely and open the Under Auger Material Cutoffs.
2. Remove the auger cover(s) bolts (4) and washers attaching auger cover to the auger chain cover. **(Figure 5-52)**

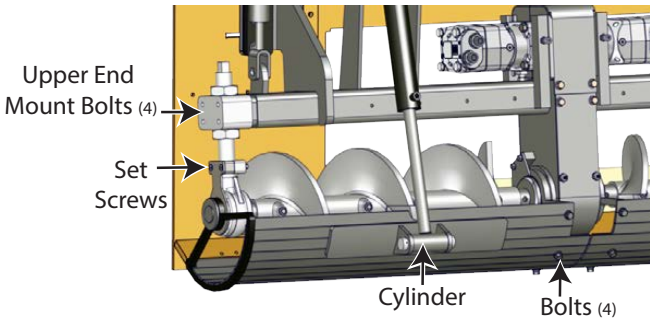


Figure 5-52. Replace Auger End Mounts

3. Remove the clevis pins holding the cylinder onto the cover, and carefully pull the cover outward away from the frame.
4. Remove the four (4) upper end mount bolts from the auger support beam.
5. Loosen the two set screws on the auger end mount collar, then slide the end mount and collar off the auger shaft.
6. Replace with new end mount, ensuring the end mount assembly is properly aligned with the auger support beam and auger shaft.
7. Retighten the auger shaft set screws and reinstall upper end mount bolts.
8. Align the cover with the auger shaft and reattach with the four bolts and torque. ***(See Section 2 for torque specifications.)***
9. Align and reinsert the clevis pins holding the cylinder in place.

Auger Inner Bearings

In time, the auger and inner bearings may become worn from use and need replacement. **(Figure 5-53)**

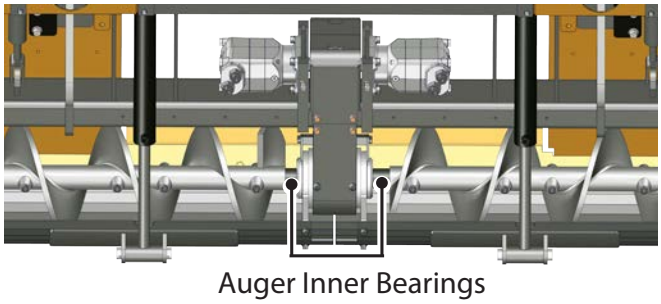


Figure 5-53. Auger Inner Bearings

To replace the auger inner bearings:

1. Extend the screed extensions completely and open the Under Auger Material Cutoffs. You will need to swing out the operator platforms for space to work.
2. Loosen the four (4) bolts on each auger motor mounting bracket. **(Figure 5-54)**

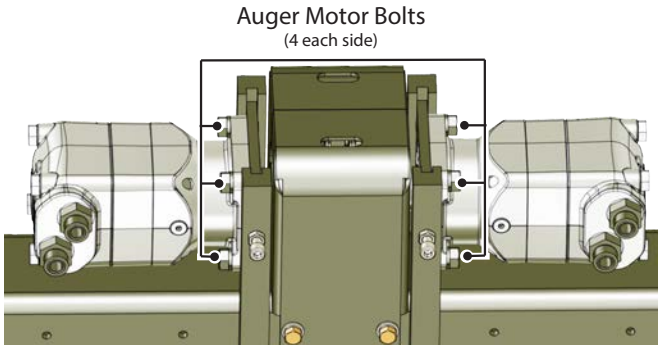


Figure 5-54. Auger Motor Bolts

3. Loosen the two (2) drive chain tensioners. **(Figure 5-51 *on the previous page.)*** The auger motors should slide down as you loosen the tensioners (tap with a rubber mallet if needed).
4. Remove the four (4) auger end mount bolts on each side of the auger, then remove end mount. **(Figure 5-52)**
5. Remove the auger inner shaft bolts. **(Figure 5-55)**

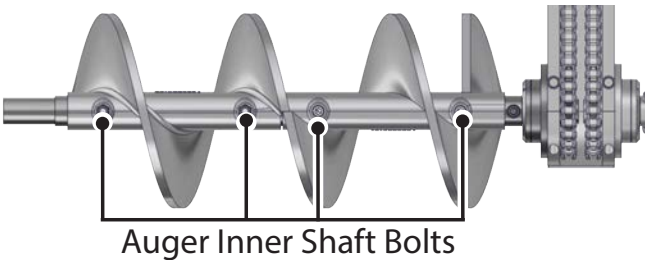


Figure 5-55. Auger Inner Shaft Bolts

6. Carefully remove the auger.
7. Remove the four (4) bolts from the inner bearing, then remove bearing and replace with new one. **(Figure 5-56)**

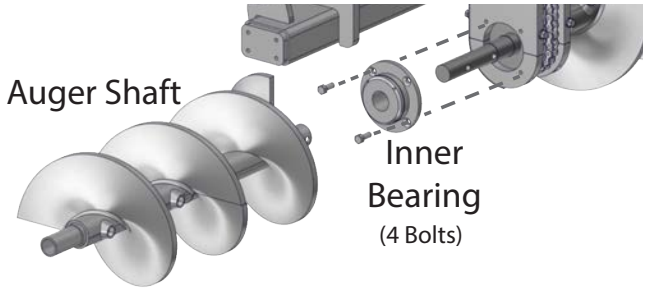


Figure 5-56. Remove Auger Inner Bearing

8. Reach in between the front of the auger and paver frame to remove the auger motor cover. Then remove the six (6) bolts holding the auger support cover in place and remove. **(Figure 5-50 *on Page 5-43.)***
   - You may need to get under the paver if you can't reach the lower two bolts on the auger support cover.
9. Remove the auger inner bearing clamps holding inner bearings into place.
10. Align new augers the same as removed augers, ensuring they are on correct side to auger material outward.
11. Reinstall auger components in reverse order.
    - Feed auger chains back onto the sprocket, ensuring chains have approximately 1/4" of slack.
    - Adjust the auger motors up by tightening the drive chain tensioners, then tighten the motor mount bolts and torque. ***(See Torque Specifications in Section 2.)***
    - Lubricate chains.
12. Operate augers to ensure proper function.

NOTE: Auger chains should be lubricated each day by spraying oil or chain lube through the slots where the auger motor is adjusted.

Main Screed Wear Plate

The screed wear plates can wear out over time and must be replaced if the plate has worn to less than 1/4-inch thick. Poor maintenance, paver cleaning, asphalt mixes containing hard aggregate and improper material flow can contribute to excessive wear on the main screed and extension wear plates. Correct screed heating also helps prevent excessive material build-up. An uneven, worn screed plate will produce an inferior mat

1. Retract screed extensions completely.
2. Set the hydraulic tow points and crown/valley to 0 (center).
3. Lower screed onto set-up blocks and null with flight screws.
4. Remove quick-release latch and then bolts holding rear walk boards (both sides). **(Figure 5-57)**

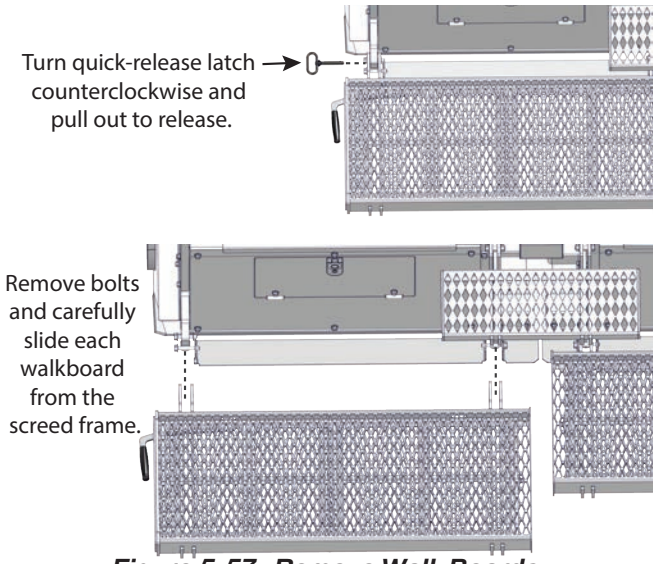


Figure 5-57. Remove Walk Boards

5

5. Remove the four (bolts) bolts holding the first screed stair and remove stair. **(Figure 5-58)**

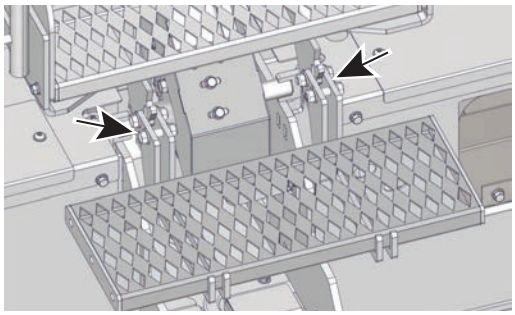


Figure 5-58. Remove First Stair

6. Remove bolts, nuts and washers on each screed cover. Remove main screed covers and disconnect wiring. **(Figure 5-59)**

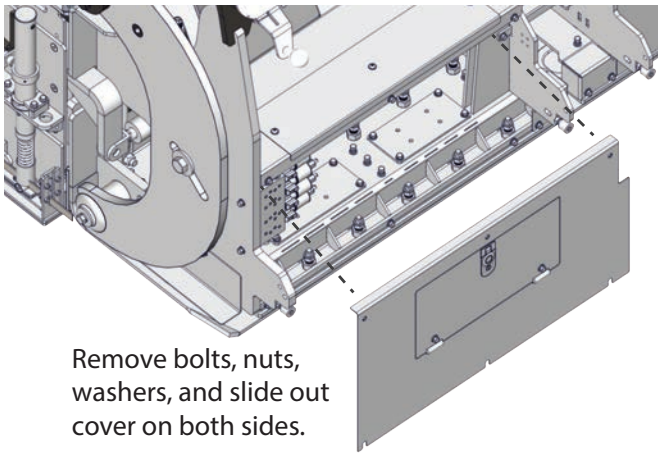


Figure 5-59. Remove Main Screed Covers

7. Install the screed lock tool (provided with your paver) between the screed frames to hold them together. **Clamp the center portion of the screed frame to hold the frame in place before lifting it from the worn wear plate. (Figure 5-60)**



Figure 5-60. Screed Lock Tool

8. Remove both rows of nuts holding the wear plate to the bottom of each side of the screed. **(Figure 5-61)**

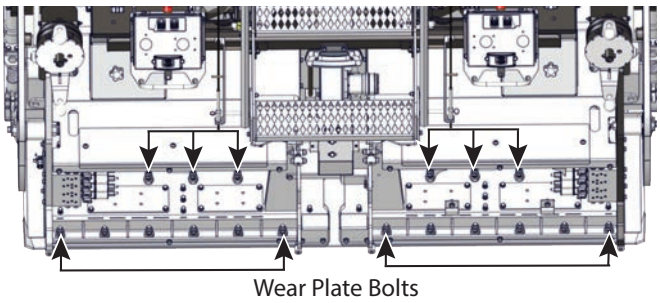


Figure 5-61. Wear Plate Bolts

9. Raise the screed completely.
10. Slide the worn wear plate from under the screed.
11. Clean all material buildup on the screed frame.
12. Set new wear plate level on three blocks, placing one block in the center and one at each end.
13. Lower the screed frame down onto the new wear plate.
14. Reinstall lock nuts and washers, but do not tighten nuts until all have been installed.
15. Adjust screed so the front bullnose of the wear plate is flat.
16. Adjust crown so the rear of the wear plate is flat. Use rear adjusters to ensure rear of wear plate is flat.
17. Reinstall the screed covers in reverse order, then the walk boards and stair.

Screed Extension Heat Box

1. Extend the screed completely.
2. Remove the sonic auger sensor if connected.
3. Remove the end gate by removing the six (6) cap screws holding the end gate. The end gate will tilt forward and slide off. **(Figure 5-62)**

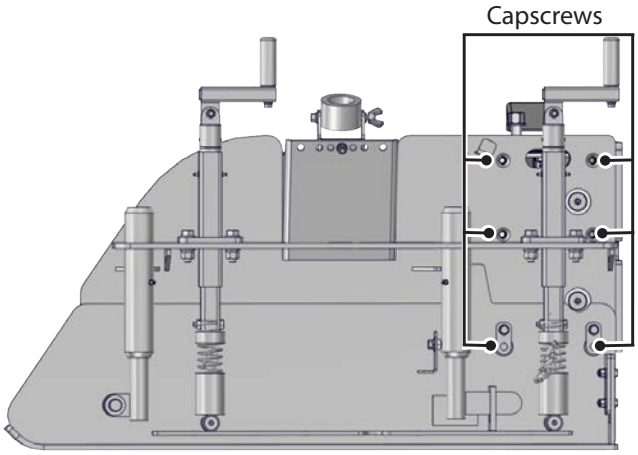


Figure 5-62. Remove End Gate

4. Disconnect the extension AOA adjuster **(Figure 5-63)** on top of the heat box by removing locknut, washer and shoulder bolt. **(Figure 5-63)**
5. Carefully slide out the hinge pin. The heat box assembly will disconnect and fall to the ground.
6. Remove the bolts and nuts holding the heat box cover, then remove.
7. Remove internal components.
8. Hold the new heat box in position and slide the hinge pin back into place.
9. Fasten the AOA adjuster back to the heat box with locknut, washer and shoulder bolt.
10. Reattach wiring.
11. Reinstall the end gate.

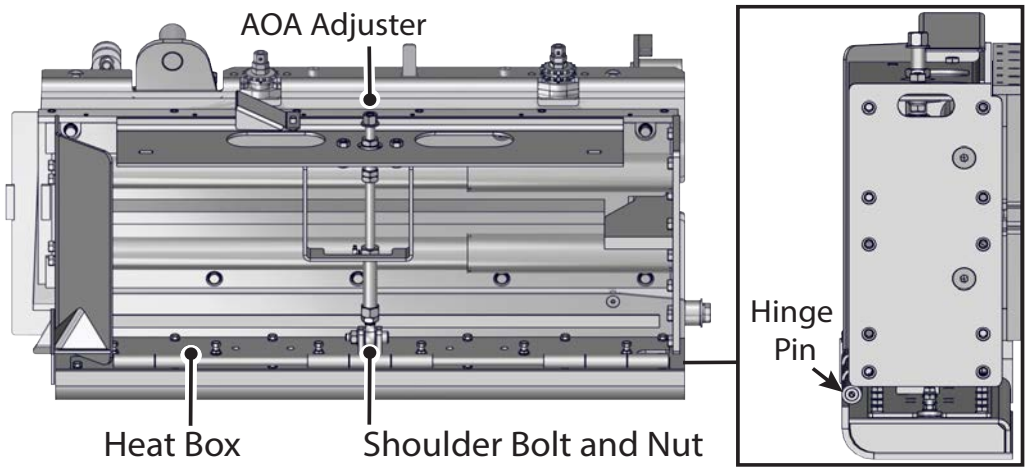


Figure 5-63. Extension Heat Box Replacement



WELDING

If the machine has been damaged or metal parts worn due to extensive use, minor welding may be necessary for repair. Welding is very dangerous. Only individuals who are properly trained should weld in a dedicated area away from others as even looking at ultraviolet light from the flame can cause serious eye injury.

Welding hazards include toxic fumes, harmful dust, light radiation, burns, combustion from sparks and the potential for extensive damage to the eye. There is also a danger of electric shock. If combustible or flammable materials are nearby, the heat and sparks produced by welding can cause fires or explosions.

TURN THE BATTERY SWITCH OFF AND DISCONNECT BATTERY TERMINALS BEFORE WELDING ON THE MACHINE.

Electrical current can ignite fire or cause an explosion, therefore it is critical to turn off power to the machine. It is also important to grind off the paint to ensure proper ground on the area being welded.

DO NOT MAKE MODIFICATIONS TO THE MACHINE. ONLY weld if needed for repair purposes. Weldment modifications to the machine can void your warranty.

Before welding on the machine:

PREPARE:

- Clear the area of any fire hazards, flammable liquids and solids, or any items that could catch on fire.
- Ensure a fire extinguisher is available in the work area.
- Hang welding blanket(s) around and underneath the work area.
- Paint and rust should be removed from the welding area to prevent arcing in other areas or components. An indirect ground can cause arcing that will damage machine components. Only connect the ground to the component being welded. DO NOT connect the ground to other bolted assemblies, wheels, bearings, motors, tracks, torque hubs, etc.

DISCONNECT:

- Turn the master battery switch to the OFF position.
- Disconnect the battery positive and negative terminals.
- Disconnect Plus 1 controller(s).
- Disconnect engine ECU(s) controller (if equipped).
- Disconnect engine DEF controller (if equipped).
- If machine has truck chassis: Disconnect the battery positive and negative terminals and unplug the ECU(s) for the engine, transmission and ABS systems if equipped. Refer to the truck owner manual for any other recommendations.

WINTERIZING FOR STORAGE

Perform the following procedures to prepare the machine for winterizing and storage.

1. Store the machine in a dry, protected area:
   - Cover seat with plastic. Place a "DO NOT OPERATE" tag on the main control panel.
   - Cover all exposed control panels with tarps or plastic.
   - Ensure all cab doors and windows are closed and seal properly.
   - Ensure the tires are properly inflated.
   - Replace worn or damaged decals if needed.
2. Inspect the machine thoroughly and repair worn or damaged parts. Inspect all hoses, couplers, fittings and cylinders. Tighten loose fittings and replace any worn hoses found.
3. Lubricate all grease points:
   - Coat all exposed cylinder shafts with lithium-based grease.
   - Ensure all exposed threads are lubricated.
   - Ensure cables are lubricated and move freely.
   - If the machine has high- and low-speed chains, be sure they are oiled.
4. Check air filter. Clean or replace air filter elements if needed.
5. Check engine oil level and fill if needed.
   - Drain dirty engine oil, fill with new oil and run the engine five minutes to allow the oil to penetrate components.

⚠WARNING To avoid the danger of exhaust fume poisoning, DO NOT operate the engine in a closed building without proper ventilation. Check hydraulic oil sight gauge and fill if needed.

NOTICE **Hydraulic oil that is contaminated must be drained, the filter elements replaced and the hydraulic tank refilled. Failure to perform this maintenance procedure if needed could result in premature failure of the pumps or motors.**

6. Ensure the fuel tank is full to prevent condensation in the fuel tank. Add cold-weather additive when appropriate.
7. If machine is equipped with Diesel Exhaust Fluid (DEF), the DEF may stored in the machine's tank up to four months. If storing any longer, drain the DEF fluid into an approved container.

NOTICE **Failure to drain the DEF tank if storing for extended periods may result in fluid deterioration that can cause DEF sensor damage. Replace the DEF fluid before operating to prevent SCR system malfunction and/or damage.**

8. If machine is equipped with a citrus tank, ensure the tank is full to prevent rust.
9. If machine is equipped with water tank(s), the tank should be drained. The pump and filter must also be purged to prevent freezing.
10. If machine is equipped with windshield washer bottle, ensure the bottle is full of window washer fluid with a cold-weather blend.
11. Perform specific gravity test on engine coolant. Drain and replace or fill with antifreeze mixture to prevent freeze damage if needed.

KUBOTA ENGINE NOTICE: If not using anti-freeze, open the cock at the bottom of radiator and remove the pressure cap to drain water completely. Leave the cock open. Place a "No Water" sign on the pressure cap.

12. Disconnect battery, adjust the electrolyte level, and recharge it. Store battery inside in a cool, dry place. Ensure breakers are in the OFF position.
13. Remove steering control boxes and electrical cords. Store inside in a cool, dry place.
    - If equipped with any plugged-in remote sensors (i.e., sonic auger sensor), disconnect, remove and store inside in a cool dry place.
14. Cover the exhaust and intake filter. Be sure to seal the exhaust cover with tape to prevent moisture from entering, especially if storing outside.
15. If the machine is equipped with a propane system, remove the propane gas bottle and store in a safe area away from the machine. Purge the gas system and seal connection(s).
16. Lower machine components and attachments (i.e., screed, broom, blade, etc.) to the lowest position, resting on blocks to prevent ground contact or pavement damage.
17. Clean the machine, including the undercarriage. Paint chipped or rusty areas to prevent further rusting. Lock all lockable panels, cab doors and spare tire. Remove ignition key and store keys in a safe place.
18. If storing machine for an extended time, run the engine five minutes every two to three months (under no load) to keep engine free from rust caused by moisture in the air. If you do not run the engine for six months or longer, apply engine oil to the valve guide and stem seal before starting.

TROUBLESHOOTING

Hydraulic Pressure

The hydraulic pressure screen on the DM430 digital controller provides an easy and convenient means of checking pressure for the five main functions of the paver: Forward propel (left and right sides), conveyors, cylinders, and augers. **(Figure 5-61)**

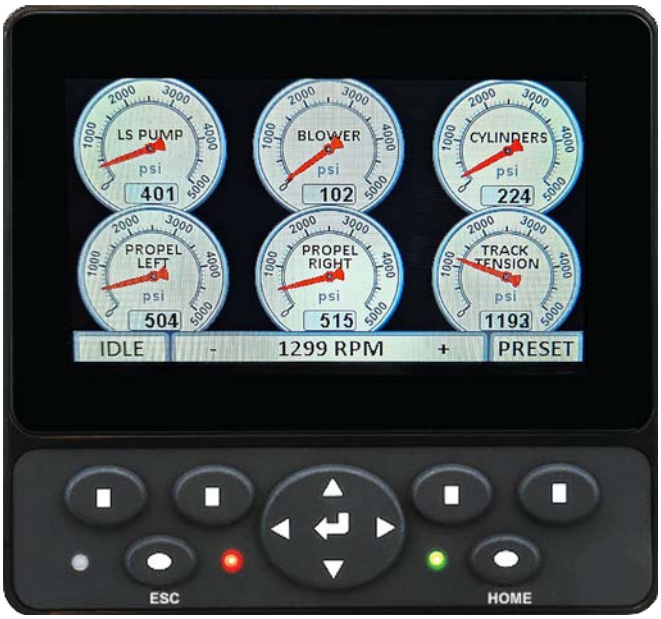


Figure 5-64. Pressure Gauge Screen

Relief pressure PSI for the paver functions are shown in the following table:

Table 5-4. Relief PSI Chart

ITEM	SPECIFICATION
Drive	5510 psi
Conveyors/Augers (Load Sense)	3480 psi
Cylinders	3000 psi
Track Tension Pressure	1200 psi
Track Tension Relieving	3600 psi
Standby Pressure	350 psi
Charge Pressure	500 psi

Track Tension Pressure Relief

The hydraulic adjustment cylinders are automatic and provide even tension on the tracks to prevent excessive wear to the paver's undercarriage. The track tension main manifold is mounted on the side of the lower manifold on the left side of the paver. There are two relief cartridge vales that function independently with ports for the left track (LT) and right track (RT). **(Figure 5-62)**

The track hydraulic pressure is set at 900 PSI at the track tension main manifold.

To check the relief pressure, perform the following procedure:

1. Connect a 3000 PSI gauge in-line with either Port LT or RT on the track tension main manifold.
2. Place a Porta Power between the front idler and frame to release pressure on the idler.
3. Increase the Porta Power pressure until the front idler cylinder compresses.
4. Pressure should read 3600 PSI.
5. If the pressure is not correct, remove cap on the relief cartridge valve. Using an allen wrench, adjust the relief cartridge valve IN for more pressure or OUT for less pressure until the PSI gauge reads 3600 PSI.
6. When the pressure gauge reads 3600 PSI, disconnect Porta Power and gauge.

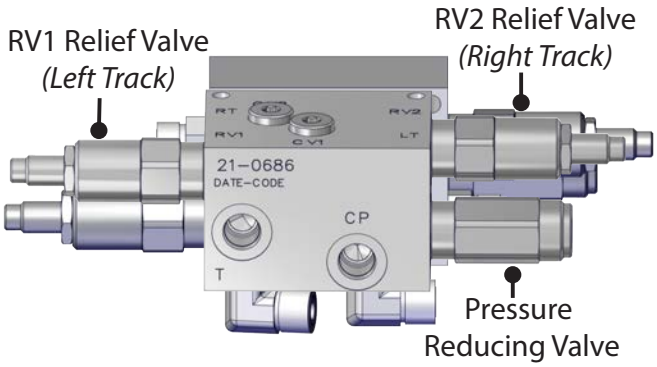


Figure 5-65. Track Tension Main Manifold

Spray Down Pump

The following troubleshooting information is provided in the event of any issues with the spray down pump (located inside the left engine access door). **(Figure 5-63)**



Figure 5-66. Spray Down Pump Location

An electronic pressure sensor displays solid or flashing LED lights on top of the pressure switch for pump status notifications. **(Figure 5-62)**

Green:

- Solid green means the pump is spraying.
- Flashing green means the pump is in a stand-by mode, ready to spray.

Yellow:

- Solid yellow means the pump is priming.

Yellow/Blue:

- Flashing yellow and blue means the pump has incorrect voltage.

Red/Blue:

- Flashing red and blue means the pump has timed-out without fluid. Cycle pump power to reset.

Red:

- Solid or flashing red light means there is an error. Consult owner's manual for troubleshooting.

NOTE: If an overload occurs, the pump stops and a yellow-green LED will flash. The electronic system will try to reactivate the pump at preset intervals.

If the overload reaches a dangerous level, the red LED will become solid and you must switch the power off and start the pump again.

NOTE: When starting the spray down pump or emptying the tank, open the bleeder valve slightly to expel air and facilitate priming. (Figure 5-64) As soon as the pump is operating, close the bleeder valve.

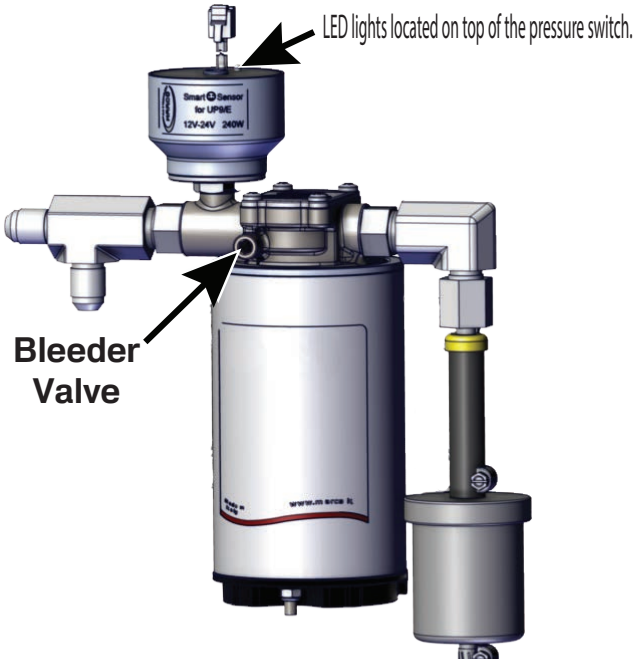


Figure 5-67. Spray Down Pump

If the spray down pump has stopped or will not start:

1. Check the effectiveness of the battery power supply (voltage activity).
2. Check that the fuse has not blown.
3. Check for any foreign material present in the hose.
4. Avoid running the pump dry for more than a few minutes. (Pumps run dry by the absence of fluid are not covered by warranty.)
5. If the pump does not prime itself:
   - Raise the screed completely, ensuring citrus tank on the screed is filled.
   - Loosen the bleeder valve inside of the cut-out until the pump primes.
   - Retighten the bleeder valve.

Conveyor Chain Tension Slack Adjustment

The conveyor chains will need periodic adjustment. Generally, the slack of a roller chain should be on the lower side. **(Figure 5-65)**

Adequate slack (S) is about four (4) percent of the span for normal drives, but the slack should be about two (2) percent of the span for the auger chains:

- Vertical drive or close to vertical drive.
- Center distance between two shafts is greater than three (3) feet.
- Chain is operated under heavy load and high frequency of on and off drive.
- Direction of the drive is often changed.

⚠CAUTION **Conveyor chains must have a slack of 3/8 to 1/2 inch. To measure deflection, exert 30 pounds of force to chain in either direction.**

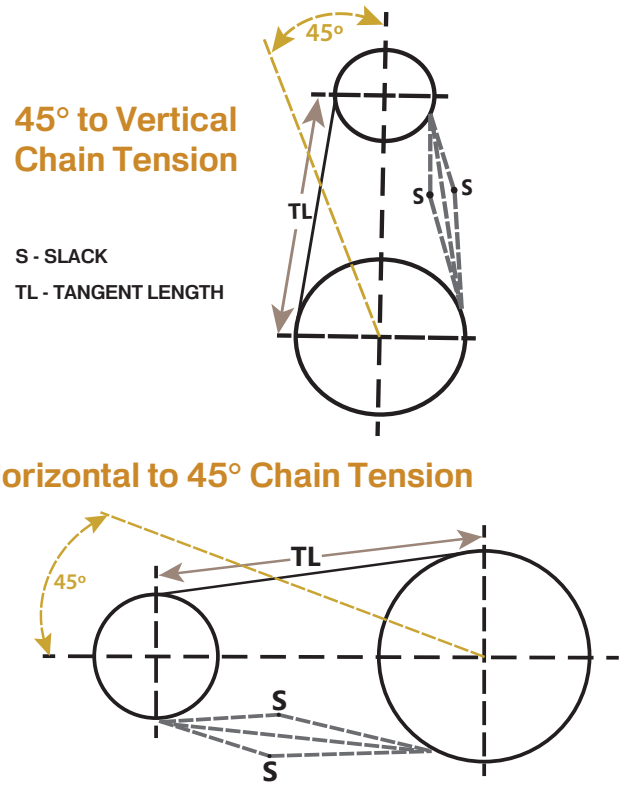


Figure 5-68. Chain Slack Adjustment for Proper Tension

REQUIRED MID-SPAN MOVEMENT

	Tangent Length (TL) Between Sprockets in Inches (Centimeters)					
	10 (25)	20 (51)	30 (76)	50 (127)	70 (178)	100 (254)
	Tangent Length Between Sprockets in Inches (Millimeters)					
Horizontal To 45°	0.4 - 0.6 (10 - 15)	0.8 - 1.2 (20 - 30)	1.2 - 1.8 (30 - 46)	2.0 - 3.0 (51 - 76)	2.8 - 4.2 (71 - 107)	4.0 - 6.0 (102 - 152)
45° To Vertical	0.2 - 0.3 (5 - 8)	0.4 - 0.6 (10 - 15)	0.6 - 0.9 (15 - 23)	1.0 - 1.5 (25 - 38)	1.4 - 2.1 (36 - 53)	2.0 - 3.0 (51 - 76)

TROUBLESHOOTING CHART

The troubleshooting chart below identifies the most common symptoms of mechanical problems. Fault codes are also shown on the DM430 digital display unit on the operator control panel consoles.

Contact your authorized LeeBoy dealer for assistance, service or repair.

Table 5-1. Paver Troubleshooting

SYMPTOM	CAUSE	REMEDY
Engine does not start.	Defective battery or low battery charge.	Replace or charge battery as applicable.
	Master switch is not set at the ON position.	Set switch to the ON position.
	Insufficient fuel supply.	Fill fuel tank.
	Safety switch is faulty.	Replace.
	Solenoid doesn't function properly.	Ensure wires are tight. Clean solenoid plunger. Replace solenoid coil. Replace or rebuild starter or solenoid.
	Switch box unplugged.	Replug switch box.
	Neutral switch defective.	Replace.
	Engine starter relay faulty.	Replace.
	E-Stop engaged.	Disengage E-Stop.
Engine turns over but does not start.	Low fuel.	Add fuel to fuel tank.
Engine is difficult to start.	Fuel is thick and doesn't flow.	Check the fuel tank and remove water, dirt or other impurities. Check the fuel filter cartridge and replace if necessary.
	Engine oil becomes thick in cold weather.	Use oil of different viscosities suitable to ambient temperatures. (Use 10W-30 in winter.)
	Battery.	Charge battery.
Engine suddenly stops.	Lack of fuel.	Refuel. Check the fuel system. Bleed the fuel system if necessary.
	Moving parts overheated.	Contact your authorized dealer.
	E-Stop button is engaged.	Check E-Stop button setting.
	Forced stop due to an engine issue.	The ECU may have forced the engine to a stop. Check the engine warning shown on the DM430 digital display and/or Diagnostic Trouble Code. If you cannot repair the problem, contact your authorized dealer.

SYMPTOM	CAUSE	REMEDY
Engine overheats.	Low oil.	Check oil level. Add oil if needed.
	Fan belt stretched, damaged, or broken.	Adjust belt tension or replace.
	Coolant low.	Add more coolant.
	Excessive concentration of antifreeze.	Add water only. Coolant should always be mixed 50/50 of coolant and water.
	Radiator net or fin clogged with dust and dirt.	Clean or replace defective parts.
	Thermostat defective.	Check thermostat and replace if needed.
	Defective temperature gauge or sensor.	Check temperature with a thermometer. Replace gauge or sensor if needed.
	Overload running.	Reduce load.
	Head gasket leaking.	Replace gasket.
Low battery.	Faulty alternator.	Replace or rebuild.
	Indicator light, bad bulb.	Replace.
Machine will not move.	RUN/STOP switch faulty.	Check RUN/STOP switch.
Machine does not steer straight.	One of the hydraulic drive motors needs adjustment.	Readjust motor(s).
	Steering control is not centered.	Center steering control.
	Travel pump defective.	Replace or rebuild travel pump.
Machine does not change speed when the 2-Speed switch is engaged.	Defective relay.	Replace relay.
	Defective solenoid.	Replace solenoid.
	Defective drive motor.	Replace drive motor.
Conveyor does not function on one or both sides.	Defective Automatic/Manual switch.	Replace switch.
	Solenoid defective.	Replace.
	Feeder drive chain broken.	Repair or replace chain.
	Defective conveyor motor.	Replace.
	Rear conveyor shaft broken.	Replace.
Tracks don't operate smoothly.	Tracks too loose.	Tighten track pad bolts.
	Track rollers worn.	Replace.
	Track tension pressure.	Check tension pressure and relieve or adjust.
Machine will not pull on one or both sides.	Faulty hydraulic motor.	Replace.
	Pump pressure too low.	Check pump pressure and relieve or adjust.
	Faulty torque hub.	Rebuild or replace.
Engine runs but there is no hydraulic pressure.	Pump drive coupling faulty.	Replace.
	Defective pump.	Replace.

SYMPTOM	CAUSE	REMEDY
Auger not turning properly.	Drive chain is too loose.	Adjust.
	Drive chain broken.	Replace.
	Faulty motor.	Replace.
	Solenoid valve defective.	Replace solenoid.
	Asphalt buildup.	Clean thoroughly and grease.
	Overload of material to augers.	Set conveyor to proper speed in AUTO or use paddles in MANUAL position.
Screed extensions binding.	Asphalt buildup.	Keep clean and greased.
Screed extension loose.	Out of adjustment.	Adjust hold-downs on extensions.
	Screed worn out or damaged.	Replace.
	Extensions set too low.	Adjust extension. Always begin operation with extensions extended up with no down pressure.
	Screed not heated properly.	Check elements and heat control box.
Screed leaving ripples in pavement.	Extension set too low.	Readjust extensions.
	Extensions work up and down.	Adjust top guide.
	Extension rod bushings worn.	Replace bushings.
Flight screw locking up.	Twisting screed too far.	Give screed time to react.
	Screw seized.	Replace screw.
Flight screw bearing damage.	Twisting screed too far.	Give screed time to react.
	Loading and unloading.	Check ramps for easy access.
Hydraulic oil leaking from breather cap.	Hydraulic oil tank overfilled.	Drain to proper fill level.
	Oil overheated.	Slow paver down.
		Check oil cooler and thermostat.
Hydraulic pump cavitation or lost power.	Low level in hydraulic tank.	Refill to proper level.
	Clogged filters.	Replace filters.
	Suction hose loose.	Retighten.
	Charge pump worn.	Rebuild or replace charge pump.
Feeder does not work on one or both sides.	Defective switch.	Adjust or replace switch.
	Solenoid defective.	Replace solenoid.
	Feeder drive chain broken.	Adjust or repair chain.
	Defective conveyor motor.	Replace motor.
	Rear conveyor shaft broken.	Replace conveyor shaft.
Feeder flight bars don't function properly.	Flight chains too loose.	Adjust.
	Feeder drive chain too loose.	Adjust.
Loss of power to drives.	Relief out of adjustment.	Check pressure and adjust if needed.
	Piston groups worn.	Replace.
	No charge pressure.	Check that charge pressure is 280 - 425 PSI.

SYMPTOM	CAUSE	REMEDY
Electric screed heating system will not operate.	Control box power switch not in ON position.	Ensure that the screed operator control box Power switch is ON.
	Breakers are in a “tripped” position.	Ensure all element breakers are in the proper set position.
	Generator malfunction.	Check voltage and frequency. Rebuild or replace if needed.
Only part of the electric screed heats.	Screed section not plugged into bottom of the control box.	Replug.
	Element breakers for screed section in a “tripped” condition.	Ensure the element breakers for that screed section are not “tripped.”
Electric screed heat system doesn’t start, or starts and doesn’t run very long.	Heating system timed out.	Check timing relay. Adjust or replace.
Electric Screed is heating, but never gets hot enough to pave.	Elements improperly clamped.	Ensure elements are clamped properly.
	Generator malfunction.	Ensure generator is “spinning.” Check voltage and frequency. Rebuild or replace if needed.
Elements have been tested, but the breaker still trips.	Faulty element wiring.	Inspect and test element wiring.

DISPLAY DIAGNOSTIC TROUBLE CODE (DTC) CHART

LEGEND

SPN: Suspect Parameter Number
DPF: Diesel Particulate Filter
FMI: Failure Mode Identifier
EGR: Exhaust Gas Recirculation
ECU: Engine Control Unit
CPU: Central Processing Unit
NE-G: Negative Ground
NE: Neutral
G: Ground
SCV: Selective Control Valve
MAF: Mass Air Flow
PM: Particulate Matter
DC: Direct Current
IC: Integrated Circuit
EBC: Electronic Brake Control
PLV: Pressure Limiting Valve
CAN: Controller Area Network
CM: Control Module
DDC: Direct Digital Control
SW: Switch
TSC: Torque Speed Control
ETC: Electronic Throttle Control
CCVS: Current Controlled Voltage Source
MPROP: Manifold Pressure Rail Operating Pressure
EEPROM: Electrically Erasable Programmable Read-Only Memory

SPN	FMI	DTC	SYMPTOM/CAUSE	RECOVERY FROM ERROR
29	3	Accelerator position sensor 2: High	Battery short circuit out of sensor/ harness. Voltage of accelerator position sensor 2 is 4.8V or less.	Forced idle. (Accelerator = 0%) Diagnostic counter = zero
29	4	Accelerator position sensor 2: Low	Ground short circuit or open circuit of sensor/harness. Voltage of accelerator position sensor 2 is 0.3V or less.	Forced idle. (Accelerator = 0%) Diagnostic counter = zero
91	2	Accelerator position sensor correlation error	Deviation from designed correlation in two sensors. Voltage of accelerator position sensor 1 is 4.8V or less.	Forced idle. (Accelerator = 0%) Diagnostic counter = zero
91	3	Accelerator position sensor 1: High	Battery short circuit or open circuit of sensor/harness. Voltage of accelerator position sensor 1 is 4.8V or less.	Forced idle. (Accelerator = 0%) Diagnostic counter = zero
91	4	Accelerator position sensor 1: Low	Ground short circuit or open circuit of sensor/harness. Voltage of accelerator position sensor 1 is 0.3 or less.	Forced idle. (Accelerator = 0%) Diagnostic counter = zero
100	1	Oil pressure error.	Engine oil pressure.	Key switch turn OFF.
102	3	Boost pressure sensor: High	Failure of sensor. Voltage of boost pressure sensor is 4.9V or above.	Key switch turn OFF. (Default value is set in consideration with high altitude usage. Engine power drops.)
102	4	Boost pressure sensor: Low	Failure of sensor. Voltage of boost pressure sensor is 0.2V or below.	Key switch turn OFF. (Default value is set in consideration with high altitude usage. Engine power drops.)
108	3	Barometric pressure sensor error: High	Barometric voltage 4.4V or more.	Diagnostic counter = zero (Default value is set in consideration with high altitude usage. Engine power drops.)

SPN	FMI	DTC	SYMPTOM/CAUSE	RECOVERY FROM ERROR
108	4	Barometric pressure sensor error: Low	Barometric voltage 1.6V or less.	Diagnostic counter = zero (Default value is set in consideration with high altitude usage. Engine power drops.)
110	0	Engine overheat.	Overheat of engine temperature greater than 248° F (120° C)	Diagnostic counter = zero Output limitation: approximately 75% of normal condition. EGR stop. Intake throttle 100% open.
110	3	Coolant temperature sensor: High	Open circuit or +B short circuit of sensor/harness. Voltage of coolant temperature sensor is 4.9V or above.	Key switch turn OFF. (White smoke increases at low temperature.)
110	4	Coolant temperature sensor: Low	Ground short circuit of sensor/ harness. Voltage of coolant temperature sensor is 0.1V or less.	Key switch turn OFF. (White smoke increases at low temperature.)
132	1	Intake air volume: Low	Engine inlet air mass flow rate tacking. (Disconnect turbo blower intake hose.)	Key switch turn OFF.
132	3	MAF sensor: High	Engine inlet air mass flow rate less than half of target value. MAF sensor voltage: 4.9V or more at normal operating condition.	Key switch turn OFF. **Engine is not forcibly stopped by ECU. STRONGLY recommend operator stop engine promptly.**
132	4	MAF sensor: Low	Open circuit or ground short circuit of sensor/harness. MAF sensor voltage: 0.1V or less.	Key switch turn OFF. **Engine is not forcibly stopped by ECU. STRONGLY recommend operator stop engine promptly.**
132	15	Boost pressure: Low	Boost pressure sensor output is below target level in high air flow operating conditions.	Key switch turn OFF. EGR stop. Engine power is restricted by boost pressure signal accordingly.
157	0	High rail pressure.	Actual pressure exceeds the command pressure. Actual pressure ≥ 197MPa (28600 psi or 2010 kgf/cm^2	Diagnostic counter = zero EGR stop.
157	3	Rail pressure sensor: High	Open circuit or +B short circuit of sensor/harness. Failure of sensor. Voltage or rail pressure sensor is 4.9V or above.	Key switch turn OFF. Engine running noise and white smoke increases. Engine forcibly stopped.
157	4	Rail pressure sensor: Low	Ground short circuit of sensor/ harness. Failure of sensor. Voltage of rail pressure sensor is 0.7V or less.	Key switch turn OFF. Engine running noise and white smoke increases. Engine forcibly stopped.
168	3	Battery voltage: High	Open circuit, short circuit or damage to harness. Failure of battery. ECU recognition of battery voltage is above 16V in 12V system. Faulty starting.	Key switch turn OFF. EGR stop.
168	4	Battery voltage: Low	Open circuit, short circuit or damage to harness. Failure of battery. ECU recognition of battery voltage is below 8V in 12V system. Faulty starting or engine may stop completely. (Not monitored during cranking.)	Diagnostic counter = zero EGR stop.

SPN	FMI	DTC	SYMPTOM/CAUSE	RECOVERY FROM ERROR
171	3	Intake air temperature built-in MAF sensor: High	Open circuit or +B short circuit of sensor/harness. Intake air temperature built-in MAF sensor voltage: 4.9V or more.	Diagnostic counter = zero
171	4	Intake air temperature built-in MAF sensor: Low	Ground short circuit of sensor/ harness. Intake air temperature built-in MAF sensor voltage: 0.1V or less.	Diagnostic counter = zero
172	0	Intake air temperature: High	Intake air temperature higher than 140° F (60° C).	Key switch turn OFF.
172	3	Intake air temperature error: High	Open circuit or +B short circuit of sensor/harness. Intake air temperature sensor is 4.9V or above.	Diagnostic counter = zero (White smoke increases at low temperature.)
172	4	Intake air temperature error: Low	Ground short circuit of sensor/ harness. Intake air temperature sensor is 0.05V or less.	Diagnostic counter = zero (White smoke increases at low temperature.)
174	0	Fuel temperature high.	Fuel temperature higher than 194° F (90° C).	Diagnostic counter = zero
174	3	Fuel temperature sensor: High	Open circuit or +B short circuit of sensor/harness. Voltage of temperature sensor in supply pump is 4.9V or above.	Diagnostic counter = zero
174	4	Fuel temperature sensor: Low	Ground short circuit of sensor/ harness. Voltage of temperature sensor in supply pump is 0.1V or less.	Diagnostic counter = zero
190	0	Engine speed.	Engine speed exceeds threshold speed. Engine speed ≥ 3500 min-1 (rpm).	Diagnostic counter = zero
628	2	Program memory.	Check-Sum error.	Key switch turn OFF. Engine stopped without delay.
633	7	Pressure limiter emergency open.	Combination of fuel leak and one of conditions below: 1. Rail pressure exceeds 191MPa (27700 psi or 1950 kgf/cm^2). 2. Within one second after the rail pressure goes below temperature in #1, it increases above.	Key switch turn OFF. EGR stop. (Engine speed may go down due to lack of fuel pressure.)
636	2	Crank position sensor pulse number error.	Open circuit or short circuit of sensor/ harness. Failure of sensor. Pulse rotation is not 56 teeth. Faulty starting. Engine vibration increases slightly.	Diagnostic counter = zero
636	7	NE-G phase shift. NE: Crankshaft position sensor. G: Camshaft position sensor.	Large phase shift between NE and G pulses within +/- 15 degrees. Engine hesitates at start-up.	Diagnostic counter = zero
636	8	No input of NE sensor pulse.	Open circuit or short circuit of sensor/ harness. Failure of sensor. No recognition of NE sensor pulse. Faulty starting. Engine vibration increases slightly.	Diagnostic counter = zero

SPN	FMI	DTC	SYMPTOM/CAUSE	RECOVERY FROM ERROR
651	3	Open circuit of harness/ coil in 1st cylinder injector.	Open circuit of harness or injector coil. Engine vibration increases. Injectors that have no error are operated.	Key switch turn OFF.
652	3	Open circuit of harness/ coil in 2nd cylinder injector.	Open circuit of harness or injector coil. Engine vibration increases. Injectors that have no error are operated.	Key switch turn OFF.
653	3	Open circuit of harness/ coil in 3rd cylinder injector.	Open circuit of harness or injector coil. Engine vibration increases. Injectors that have no error are operated.	Key switch turn OFF.
654	3	Open circuit of harness/ coil in 4th cylinder injector.	Open circuit of harness or injector coil. Engine vibration increases. Injectors that have no error are operated.	Key switch turn OFF.
676	0	Glow heater relay driving circuit overheat.	Glow relay coil resistance or load is too high. Faulty starting. White smoke increases.	Key switch turn OFF.
676	5	Open circuit of glow relay driving circuit.	Open circuit of harness or relay coil. Faulty starting. White smoke increases.	Key switch turn OFF.
679	7	Pressure limiter not open.	Rail pressure value is sticking or not enough engine power to open PLV forcibly.	Key switch turn OFF.
723	2	G sensor pulse number error.	After fault opening PLV, rail pressure is above 160 MPa.	Diagnostic counter = zero Engine stops.
723	8	No input of G sensor pulse.	Open circuit or short circuit of sensor/ harness. Failure of sensor. No recognition of G sensor pulse. Engine hesitates at start-up.	Diagnostic counter = zero
1077	2	Failure of CPU.	CPU fatal error. Engine stops.	Key switch turn OFF.
1239	1	Fuel leak in high-pressured fuel system.	Fuel consumption is calculated from the difference of fuel pressure of, before and after the injection. Error is detected when excess fuel consumption is found. Engine could stop.	Key switch turn OFF.
1347	3	+B short circuit of SCV.	EGR stops.	Key switch turn OFF.
1347	4	SVC drive system error.	Open circuit or ground short circuit of SCV.	Key switch turn OFF. EGR stops, engine forcibly stops later.
1347	5	Open circuit of SCV (MPROP)	Open circuit short of SVC.	Key switch turn OFF. EGR stops, engine forcibly stops later.
1347	7	SCV stuck in open position.	Actual rail pressure continuously exceeds the command rail pressure. Discharge request of supply pump goes below 0 mm^3/st and the actual rail pressure is 10 MPa for 26 seconds or more. Engine may stop.	Key switch turn OFF.
1485	2	Main relay locked in closed position.	Failure of main relay. Stays active longer than one second without command. Battery goes dead.	Diagnostic counter = zero